

FORM VI

MATHEMATICS EXTENSION 2

Examination date

Tuesday 5th August 2008

Time allowed

3 hours (plus 5 minutes reading time)

Instructions

- All eight questions may be attempted.
- All eight questions are of equal value.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection

- Write your candidate number clearly on each booklet.
- Hand in the eight questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Keep the printed examination paper and bring it to your next Mathematics lesson.

Checklist

- SGS booklets: 8 per boy. A total of 750 booklets should be sufficient.
- Candidature: 74 boys.

Examiner

BDD

QUESTION ONE (15 marks) Use a separate writing booklet.

Marks

- (a) By completing the square, find

2

$$\int \frac{1}{\sqrt{5-4x-x^2}} dx.$$

- (b) Use integration by parts to find

2

$$\int x e^{-x} dx.$$

- (c) Use log laws to assist in finding

2

$$\int \frac{\ln x^2}{x} dx.$$

- (d) Use the substitution $u = \tan x$ to find

3

$$\int \sec^4 x dx.$$

- (e) Use the substitution $x = 3 \cos \theta$ to find

4

$$\int \frac{x^2}{\sqrt{9-x^2}} dx.$$

- (f) Use a careful substitution and a symmetry argument to find

2

$$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (\pi - x)^2 \sin x dx.$$

QUESTION TWO (15 marks) Use a separate writing booklet.

Marks

- (a) (i) Express $1 - i\sqrt{3}$ and $1 + i\sqrt{3}$ in modulus–argument form.

1

- (ii) Hence use de Moivre’s theorem to evaluate

2

$$(1 - i\sqrt{3})^{10} + (1 + i\sqrt{3})^{10}.$$

- (b) Shade the region of the complex plane described by

3

$$|z| < 2 \quad \text{and} \quad \operatorname{Re}(z) \leq 1.$$

You need not find the coordinates of any points of intersection.

- (c) The three complex numbers $-2 + 3i$, $3 - 2i$ and $5 + 4i$ are represented by the points A , B and C respectively in the complex plane.

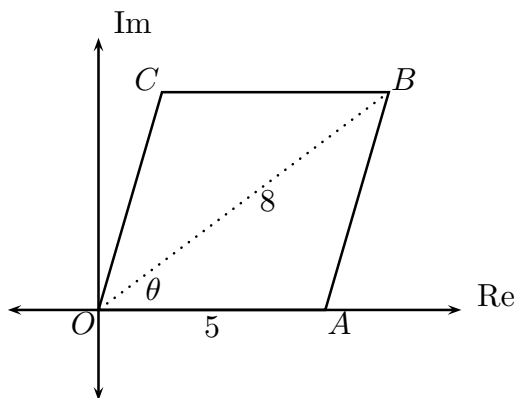
- (i) Show that $\triangle ABC$ is isosceles.

2

- (ii) Find the midpoint M of BC .

1

- (d)



The diagram above shows a rhombus $OABC$ in the first quadrant of the Argand diagram, with the origin O as one vertex and another vertex A lying on the real axis. The longer diagonal OB is 8 units, and each side is 5 units.

Let $\angle AOB = \theta$ and let $z = \cos \theta + i \sin \theta$.

- (i) Explain why OC is represented by the complex number $5z^2$.

2

- (ii) Show that z satisfies the quadratic equation $5z^2 = 8z - 5$.

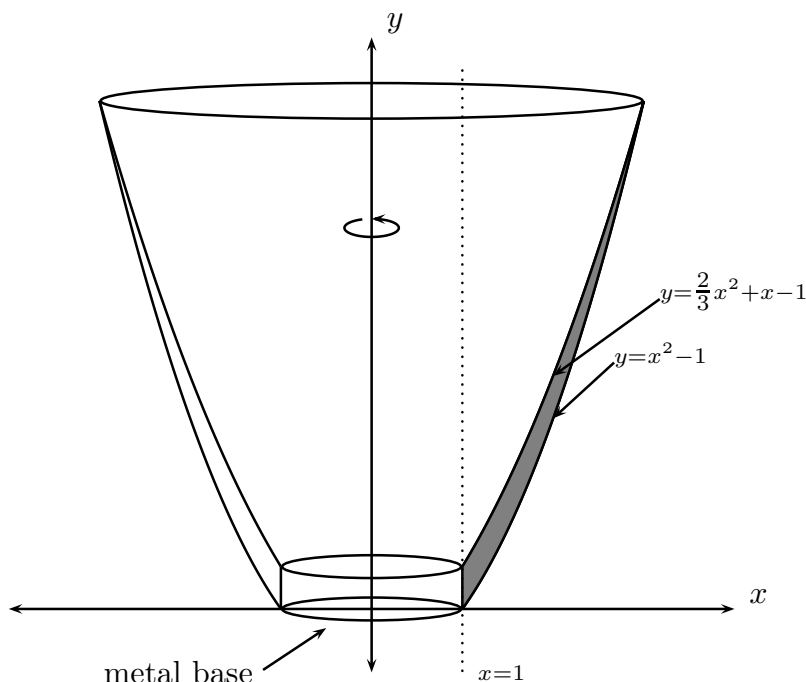
1

- (iii) Solve this quadratic equation and then find the complex numbers representing the vertices B and C .

3

QUESTION THREE (15 marks) Use a separate writing booklet.**Marks**

(a)



A glass is designed by rotating the region bounded by the curves $y = x^2 - 1$ and $y = \frac{2}{3}x^2 + x - 1$ and the line $x = 1$ about the y -axis, as in the diagram above. The resulting volume is to be grafted onto a cylindrical metal base whose volume need not be included in the following calculations.

- (i) Find the point where the curves intersect in the first quadrant. 1
- (ii) Use the method of cylindrical shells to determine the volume of glass in the solid. 3
- (b) Draw a neat half-page diagram of the ellipse $\frac{x^2}{4} + \frac{y^2}{2} = 1$ showing foci, directrices and intercepts with the axes. 4
- (c) Let $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be an ellipse with foci S and S' and eccentricity e . Carefully prove for any point $P(x_1, y_1)$ on the ellipse that $PS + PS' = 2a$. 2
- (d) Let $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos(\theta + \alpha), b \sin(\theta + \alpha))$ be two distinct points on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Let O be the origin.
 - (i) Show that line OP has equation $(b \sin \theta)x - (a \cos \theta)y = 0$. 2
 - (ii) Show that the perpendicular distance from Q to OP is 2

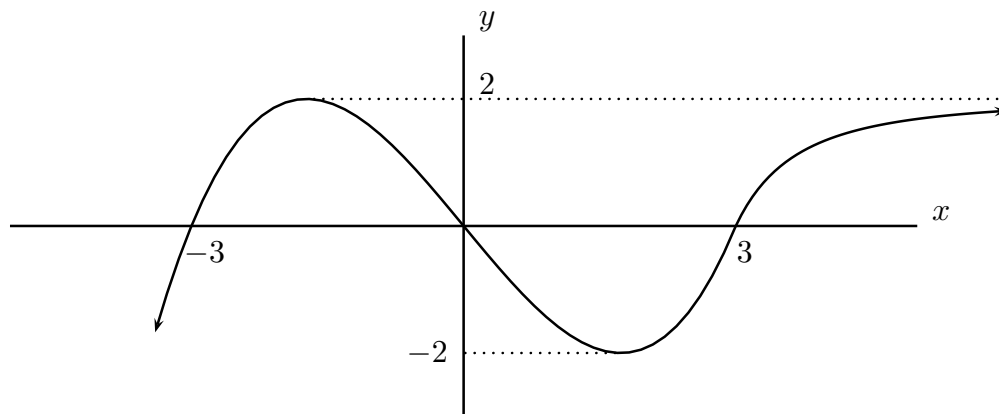
$$\frac{ab |\sin \alpha|}{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}}.$$
 - (iii) Hence show that the area of the triangle OPQ is independent of θ . 1

Exam continues next page ...

QUESTION FOUR (15 marks) Use a separate writing booklet.**Marks**

- (a) Factorise $x^5 - 1$ as the product of real linear and quadratic factors. You may leave your answer in terms of trigonometric ratios. **3**

(b)



The graph of a certain function $y = f(x)$ is sketched above. Draw neat half-page sketches of the following graphs.

(i) $y = \frac{1}{f(x)}$

2

(ii) $y = e^{f(x)}$

3

(iii) $y^2 = f(x)$

3

- (c) The line $y = 3x + 4$ is tangent to the cubic $y = 9x^3 - 48x^2 + 55x - 12$ at A , and intersects the cubic again at B .

- (i) Show that the x -coordinates of the points A and B are the roots of the cubic

1

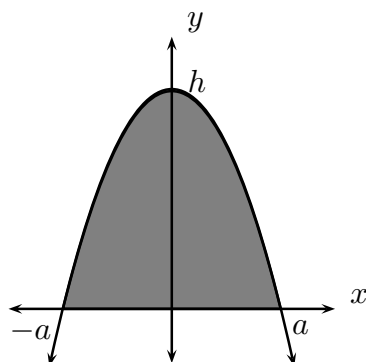
$$9x^3 - 48x^2 + 52x - 16 = 0.$$

- (ii) Explain briefly why the cubic equation in part (i) must have a double root, and use this fact to find the x -coordinates of the points A and B .

3

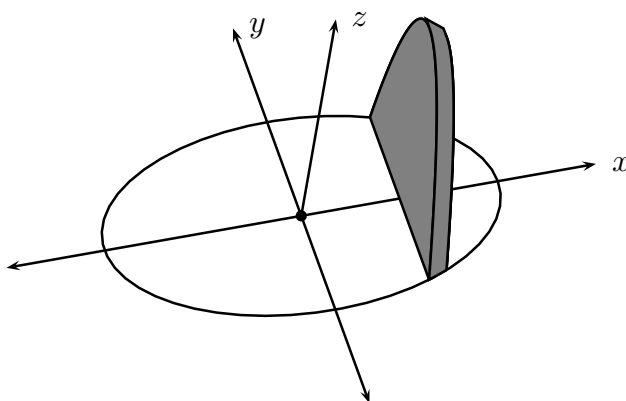
QUESTION FIVE (15 marks) Use a separate writing booklet.**Marks****3**

(a)



The diagram above shows a parabola with vertex $(0, h)$ and zeroes $x = a$ and $x = -a$. Show that the shaded area is $\frac{4}{3}ha$ square units.

(b)



A certain dome tent is designed with an elliptical base $\frac{x^2}{4} + \frac{y^2}{1} = 1$, as in the diagram above. Cross-sections perpendicular to the base are segments of a parabola. The height of each parabolic segment is equal to the width of its base.

(i) Use your result in part (a) to show that a typical cross-section has area

1

$$\frac{8 - 2x^2}{3}.$$

(ii) Show that the volume of the tent is $\frac{64}{9}$ cubic units.

2

QUESTION FIVE (Continued)

(c) (i) Expand $(\cos \theta + i \sin \theta)^7$ using the binomial theorem.

1

(ii) Use de Moivre's theorem to establish the identity

2

$$\tan 7\theta = \frac{7 \tan \theta - 35 \tan^3 \theta + 21 \tan^5 \theta - \tan^7 \theta}{1 - 21 \tan^2 \theta + 35 \tan^4 \theta - 7 \tan^6 \theta}.$$

(iii) Use your result from part (ii) to solve the polynomial equation

2

$$x^6 - 21x^4 + 35x^2 - 7 = 0.$$

(iv) Use product-of-roots to determine the value of

2

$$\tan \frac{\pi}{7} \tan \frac{2\pi}{7} \tan \frac{3\pi}{7}.$$

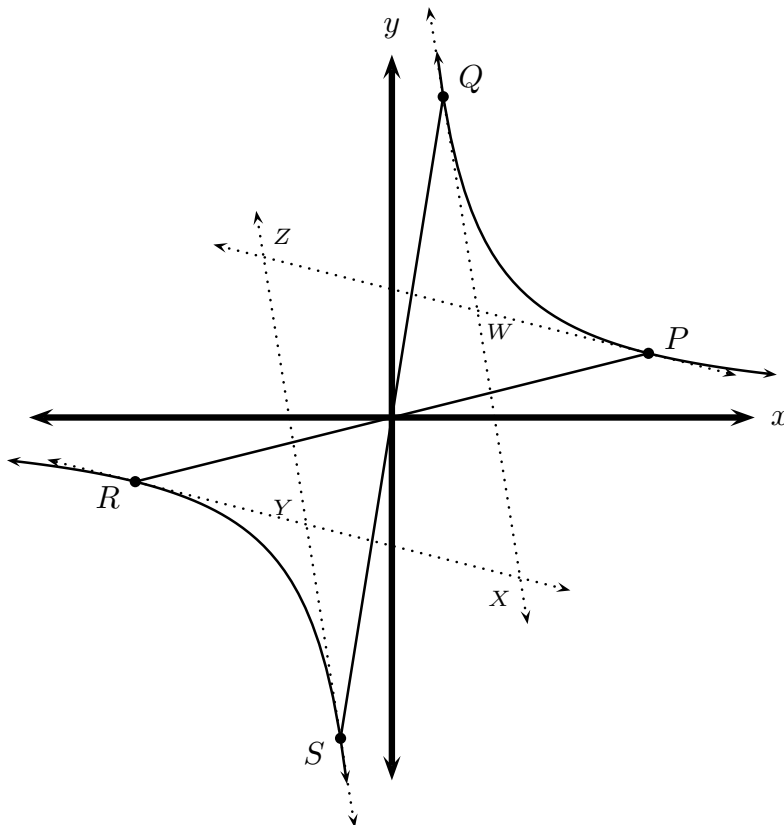
(v) Use a careful argument involving roots to determine the value of

2

$$\cot^2 \frac{\pi}{7} + \cot^2 \frac{2\pi}{7} + \cot^2 \frac{3\pi}{7}.$$

QUESTION SIX (15 marks) Use a separate writing booklet.**Marks**

(a)



Let $P\left(ct_1, \frac{c}{t_1}\right)$ and $Q\left(ct_2, \frac{c}{t_2}\right)$ be any two points on the rectangular hyperbola $xy = c^2$ with $0 < t_2 < t_1$. Let O be the origin. Let $R\left(-ct_1, -\frac{c}{t_1}\right)$ and $S\left(-ct_2, -\frac{c}{t_2}\right)$ be the points diametrically opposed to P and Q respectively. (that is, POR and QOS are straight).

- (i) Prove that P , Q , R and S are the vertices of a parallelogram. 2
- (ii) Show that the tangent at P has equation $x + t_1^2 y = 2ct_1$, and then write down the equation of the tangent at Q . 2
- (iii) Show that the point of intersection W of the tangents at P and at Q has coordinates $\left(\frac{2ct_1 t_2}{t_1 + t_2}, \frac{2c}{t_1 + t_2}\right)$. 2
- (iv) Let the tangents at Q and R meet at X , the tangents at R and S meet at Y , and the tangents at S and P meet at Z . Write down, without working, the coordinates of X , Y and Z . 1
- (v) Show that $WXYZ$ is also a parallelogram. 1

Exam continues next page ...

QUESTION SIX (Continued)

(b) Let $I_n = \int_1^2 \frac{(x-1)^{\frac{n}{2}}}{x} dx$, where n is any integer.

(i) Show that $I_{-1} = \frac{\pi}{2}$. 3

(ii) Show that $I_n = \frac{2}{n} - I_{n-2}$, for $n \neq 0$. 3

(HINT: Integration by parts is not required.)

(iii) Hence find I_5 . 1

QUESTION SEVEN (15 marks) Use a separate writing booklet.

Marks

- (a) A particle of mass m kilograms is launched vertically upwards in a highly resistive medium at a velocity 5 m/s. It is subject to the force of gravity and to a resistance due to motion of magnitude $\frac{mv^3}{100}$.

Take $g = 10 \text{ m/s}^2$ and upwards as positive.

The equation of motion is $\ddot{x} = -g - \frac{v^3}{100}$.

- (i) Show the height x above the point of launch and the velocity v are related by 1

$$\frac{dv}{dx} = \frac{v^3 + 1000}{-100v}.$$

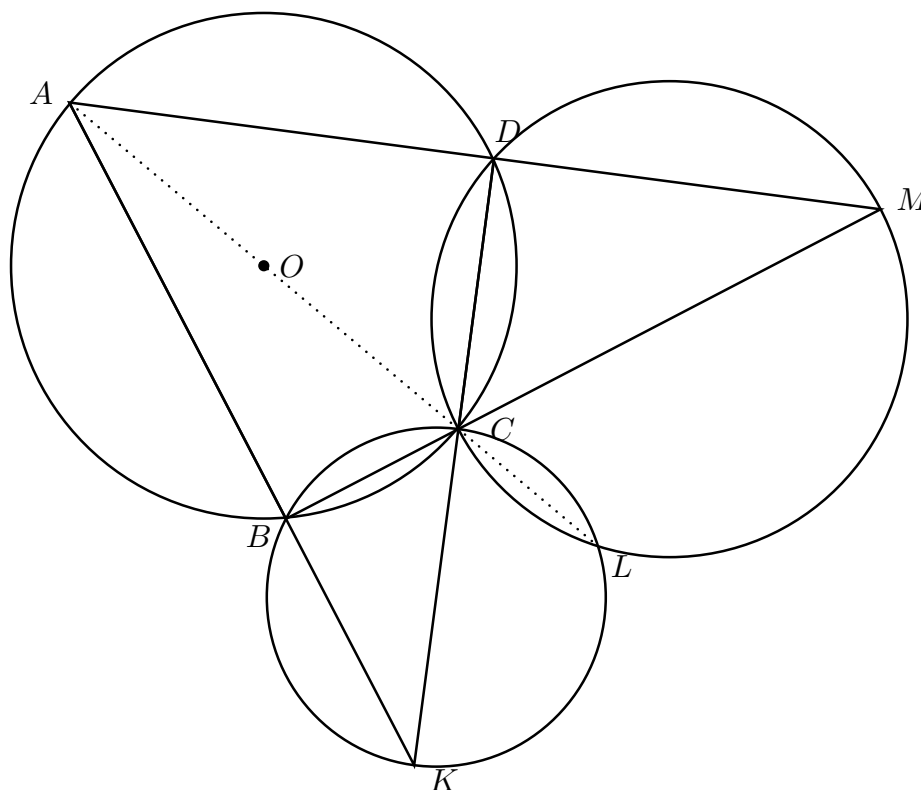
- (ii) Find constants A , B and C such that 3

$$\frac{100v}{v^3 + 1000} = \frac{A}{v + 10} + \frac{Bv + C}{v^2 - 10v + 100}.$$

- (iii) Hence find the maximum height reached by the particle, giving your answer correct to the nearest centimetre. 3

QUESTION SEVEN (Continued)

(b)



In the diagram above, the points A , B , C and D lie on a circle with centre O . The intervals AB and DC produced meet at K , and the intervals AD and BC produced meet at M . The circles BCK and DCM meet again at L .

Assume additionally that A , O , C and L are collinear.

Copy or trace the diagram into your answer book.

- (i) Prove that the centre F of the circle DCM lies on CM . 1
- (ii) Prove that K , L and M are collinear. 2
- (iii) Prove that a circle may be drawn through D , B , K and M . Label the centre of this circle X and describe its location. 2
- (iv) Prove that there is a circle passing through B , X , F and D . 3

HINT: Let $\angle DMB = \alpha$.

(The circle also passes through O , L and the centre of circle BCK , but you need not prove these facts and should not assume them.)

QUESTION EIGHT (15 marks) Use a separate writing booklet.**Marks**

- (a) (i) Show that the polynomial

1

$$P(z) = 7z^3 + (3i - 48)z^2 + 99z - 64 - i.$$

may be written as a difference of cubes:

$$P(z) = 8(z - 2)^3 - (z - i)^3.$$

- (ii) Hence solve the polynomial equation
- $P(z) = 0$
- .

4

You may wish to make use of the cube roots of unity 1, ω and ω^2 in your solution, where $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$.

Express your solutions in the form $a + ib$, where a and b are real.

- (b) Let
- x
- be any positive real number.

Define recursively two linked sequences $a_0, a_1, a_2, \dots$ and $b_0, b_1, b_2, \dots$ by

$$a_0 = (1 + x^2)^{-\frac{1}{2}},$$

$$b_0 = 1$$

$$a_{i+1} = \frac{1}{2}(a_i + b_i), \text{ for all } i \geq 0$$

$$b_{i+1} = \sqrt{a_{i+1}b_i}, \text{ for all } i \geq 0.$$

- (i) Let
- $\theta = \tan^{-1} x$
- , so that
- $x = \tan \theta$
- . Prove by induction that for all
- $n \geq 0$
- ,

4

$$\frac{a_n}{b_n} = \cos \frac{\theta}{2^n} \quad \text{and} \quad \frac{\sin \theta}{b_n} = 2^n \sin \frac{\theta}{2^n}.$$

- (ii) Prove that
- $\frac{\sin \theta}{b_n} \rightarrow \theta$
- as
- $n \rightarrow \infty$
- , and write down
- $\lim_{n \rightarrow \infty} \frac{x}{b_n \sqrt{1 + x^2}}$
- .

2

- (iii) Let
- $x = 1$
- , and evaluate three iterations to find
- a_3
- and
- b_3
- .

1

You should record each answer correct to four decimal places.

- (iv) Hence estimate
- $\frac{\pi}{4}$
- .

1

- (v) With
- $x = 1$
- , find algebraically how many iterations need to be taken before the ratio
- $\frac{b_{n+1}}{b_n}$
- differs from 1 by less than
- 10^{-10}
- .

2**END OF EXAMINATION**

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

QUESTION ONE (15 marks)

Marks

(a)

2

$$\begin{aligned}\int \frac{1}{\sqrt{5-4x-x^2}} dx &= \int \frac{1}{\sqrt{9-(x+2)^2}} dx \\ &= \sin^{-1} \frac{x+2}{3} + C\end{aligned}$$

(b)

2

$$\begin{aligned}\int x e^{-x} dx &= x(-e^{-x}) - \int 1(-e^{-x}) dx \\ &= -x e^{-x} - e^{-x} + C\end{aligned}$$

(c)

2

$$\begin{aligned}\int \frac{\ln x^2}{x} dx &= \int \frac{2 \ln x}{x} dx \\ &= \int 2u du \quad (\text{where } u = \ln x) \\ &= u^2 + C \\ &= (\ln x)^2 + C\end{aligned}$$

(d)

3

$$\begin{aligned}\int \sec^4 x dx &= \int \sec^2 x \times \sec^2 x du \\ &= \int (1+u^2) du \\ &= u + \frac{1}{3}u^3 + C \\ &= \tan x + \frac{1}{3}\tan^3 x + C\end{aligned}$$

$$\begin{aligned}\text{Let } u &= \tan x \\ \text{Then } du &= \sec^2 x dx \\ \sec^2 x &= 1 + \tan^2 x \\ &= 1 + u^2\end{aligned}$$

(e)

4

$$\begin{aligned}\int \frac{x^2}{\sqrt{9-x^2}} dx &= \int \frac{9 \cos^2 \theta}{3 \sin \theta} \times (-3 \sin \theta) d\theta \\ &= - \int 9 \cos^2 \theta d\theta \\ &= - \int -\frac{9}{2}(1 + \cos 2\theta) d\theta \\ &= -\frac{9}{2}\theta - \frac{9}{4}\sin 2\theta + C \\ &= -\frac{9}{2}\theta - \frac{9}{2}\cos \theta \sin \theta + C \\ &= -\frac{9}{2}\theta - \frac{1}{2} \times 3 \cos \theta \times 3 \sin \theta + C \\ &= -\frac{9}{2}\cos^{-1} \frac{x}{3} - \frac{1}{2}x\sqrt{9-x^2} + C\end{aligned}$$

$$\begin{aligned}\text{Let } x &= 3 \cos \theta \\ \text{Then } dx &= -3 \sin \theta d\theta \\ x^2 &= 9 \cos^2 \theta \\ 9 - x^2 &= 9 - 9 \cos^2 \theta \\ &= 9 \sin^2 \theta \\ \sqrt{9-x^2} &= 3 \sin \theta\end{aligned}$$

(f) Let $u = 2\pi - x$. Then**2**

$$\begin{aligned}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (\pi - x)^2 \sin x dx &= \int_{\frac{3\pi}{2}}^{\frac{\pi}{2}} (-\pi + u)^2 \sin(2\pi - u) (-du) \\ &= - \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (u - \pi)^2 \sin u du \\ &= - \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (\pi - x)^2 \sin x dx \quad (\text{relabelling})\end{aligned}$$

Hence the value of the integral is 0. (The substitution $u = \pi - x$ and a standard result about odd functions would also work here.)

QUESTION TWO (15 marks)

Marks

(a) (i) $1 + i\sqrt{3} = 2 \operatorname{cis} \frac{\pi}{3}$ and $1 - i\sqrt{3} = 2 \operatorname{cis} \frac{-\pi}{3}$

1

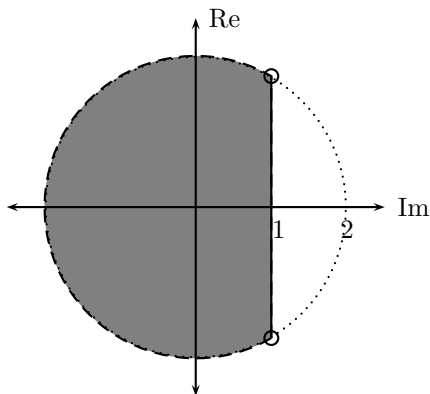
(ii)

2

$$\begin{aligned} \text{Hence } (1 - i\sqrt{3})^{10} + (1 + i\sqrt{3})^{10} &= 2^{10} \operatorname{cis} \frac{-10\pi}{3} + 2^{10} \operatorname{cis} \frac{10\pi}{3} \\ &= 2^{10} \times 2 \cos \frac{10\pi}{3} \\ &= 2^{11} \cos \frac{4\pi}{3} \\ &= -2^{10} \\ &= -1024 \end{aligned}$$

(b)

3



(c) (i) Let $a = -2 + 3i$, $b = 3 - 2i$ and $c = 5 + 4i$. Then

2

$$\begin{aligned} |a - b| &= |-5 + 5i| = \sqrt{50} \\ |a - c| &= |-7 - i| = \sqrt{50} \\ |b - c| &= |-2 - 6i| = \sqrt{40} \end{aligned}$$

The triangle is isosceles, since $AB = AC = \sqrt{50}$.

(ii) The midpoint M of BC is represented by the complex number

1

$$\begin{aligned} b + \frac{1}{2}(c - b) &= \frac{1}{2}(b + c) \\ &= 4 + i \end{aligned}$$

(d) (i) Since the diagonals of a rhombus bisect its vertex angles, $\angle BOC = \theta$. But then $\arg(OC) = 2\theta$ and $|OC| = 5$, so $OC = 5 \operatorname{cis} 2\theta = 5z^2$, by de Moivre's theorem.

2

(ii) From part (i), $OC = 5z^2$. But

1

$$\begin{aligned} OC &= OB + BC \\ &= 8 \operatorname{cis} \theta - 5 \\ &= 8z - 5 \end{aligned}$$

Hence $z^2 = 8z - 5$.

(iii) We shall use the quadratic formula to solve $z^2 - 8z + 5 = 0$.

3

The discriminant is $b^2 - 4ac = -36$ and so

$$\begin{aligned} z &= \frac{8+6i}{10} & z &= \frac{8-6i}{10} \\ &= \frac{4+3i}{5} & \text{or} & \\ & & &= \frac{4-3i}{5} \end{aligned}$$

But notice that z lies in quadrant 1, so $z = \frac{4+3i}{5}$.

Hence $OB = 8z = \frac{32+24i}{5}$ and $OC = 8z - 5 = \frac{7+24i}{5}$.

QUESTION THREE (15 marks)

Marks

(a) (i) When they intersect,

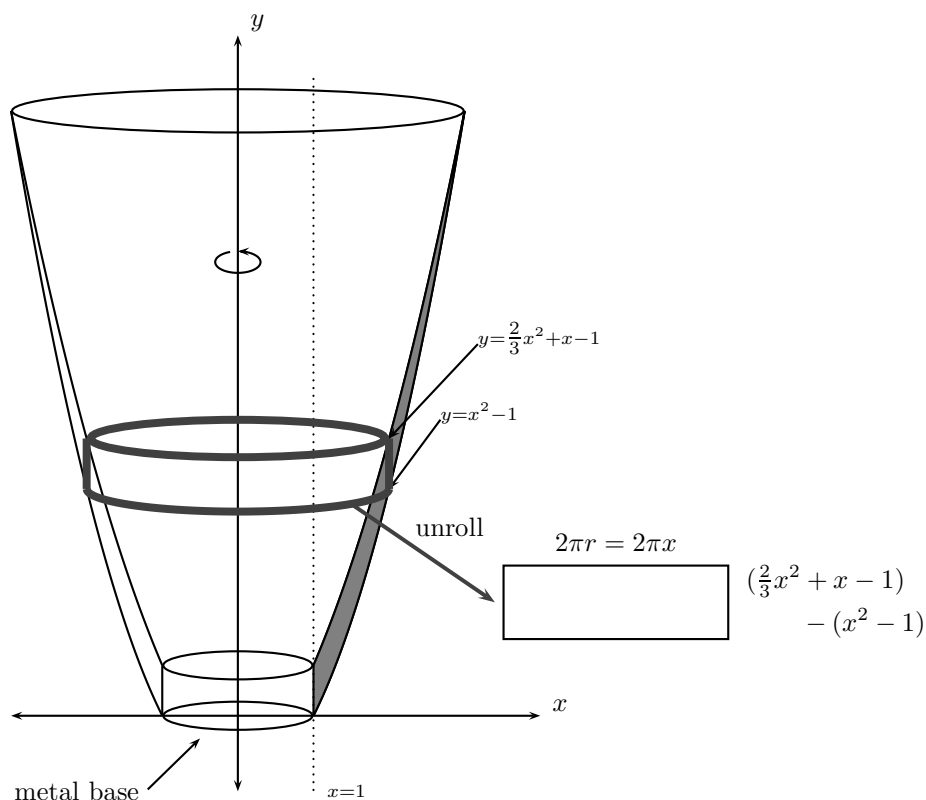
1

$$\begin{aligned} x^2 - 1 &= \frac{2}{3}x^2 + x - 1 \\ 3x^2 - 3 &= 2x^2 + 3x - 3 \\ x^2 - 3x &= 0 \\ x(x - 3) &= 0 \\ x &= 3 \quad (x > 1) \end{aligned}$$

So they intersect at $(3, 8)$

(ii)

3

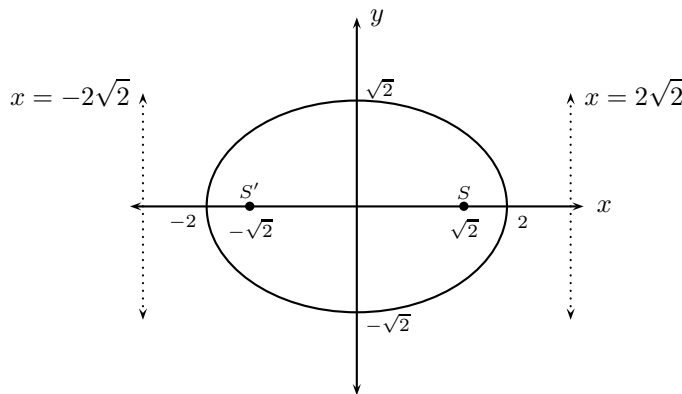


The volume of a cylindrical slice is $\delta V = 2\pi x \times \left(-\frac{1}{3}x^2 + x\right)\delta x$. Hence the total volume required is

$$\begin{aligned} V &= \lim_{\delta x \rightarrow 0} \sum_{x=1}^3 2\pi x \left(-\frac{1}{3}x^2 + x\right) \delta x \\ &= \frac{2\pi}{3} \int_1^3 (-x^3 + 3x^2) dx \\ &= \frac{2\pi}{3} \left[-\frac{1}{4}x^4 + x^3\right]_1^3 \\ &= \frac{2\pi}{3} \left(\frac{27}{4} - \frac{3}{4}\right) \\ &= 4\pi \text{ cubic units} \end{aligned}$$

(b)

4



(c) Let ℓ and ℓ' be the directrices corresponding to the focus S and S' respectively and let e be the eccentricity. Then using the eccentricity definition of a conic,

2

$$\begin{aligned} PS + PS' &= ePl + eP\ell' \quad (\text{Using the notation } P\ell \text{ for the distance from } P \text{ to } \ell \text{ etc}) \\ &= e(P\ell + P\ell') \\ &= e \times \text{distance between directrices} \\ &= e \times 2 \times \frac{a}{e} \\ &= 2a \end{aligned}$$

(d) (i)

2

$$\begin{aligned} \text{gradient } OP &= \frac{\text{rise}}{\text{run}} \\ &= \frac{b \sin \theta - 0}{a \cos \theta - 0} \\ &= \frac{b \sin \theta}{a \cos \theta} \end{aligned}$$

Using the point-gradient form $y - y_1 = m(x - x_1)$ gives

$$\begin{aligned} y - 0 &= \frac{b \sin \theta}{a \cos \theta} (x - 0) \\ a \cos \theta y &= b \sin \theta x \\ b \sin \theta x - a \cos \theta y &= 0 \end{aligned}$$

(ii) Using the perpendicular distance formula, the perpendicular distance from OP to Q is

2

$$\begin{aligned}
& \frac{\left| b \sin \theta (a \cos(\theta + \alpha)) - a \cos \theta (b \sin(\theta + \alpha)) \right|}{\sqrt{(a \cos \theta)^2 + (b \sin \theta)^2}} \\
&= \frac{ab \left| \sin \theta \cos(\theta + \alpha) - \cos \theta \sin(\theta + \alpha) \right|}{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}} \\
&= \frac{ab \left| \sin(\theta - (\theta + \alpha)) \right|}{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}} \\
&= \frac{ab \left| \sin \alpha \right|}{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}}
\end{aligned}$$

(iii) The area of $\triangle OPQ$ is

1

$$\begin{aligned}
& \frac{1}{2} \frac{ab \left| \sin \alpha \right|}{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}} \times OP \\
&= \frac{1}{2} \times \frac{ab \left| \sin \alpha \right|}{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}} \times \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} \\
&= \frac{1}{2} ab \left| \sin \alpha \right|
\end{aligned}$$

which is independent of θ .**QUESTION FOUR** (15 marks)

Marks

(a) First we need to solve $x^5 = 1$. Let $x = \text{cis } \theta$ and then by de Moivre's theorem

3

$$\text{cis } 5\theta = \text{cis } 2k\pi \quad \text{for } k \in \mathbf{Z}$$

Hence $5\theta = 2k\pi$

$$\theta = \frac{2}{5}k\pi$$

As our five roots we shall take $\text{cis } 0 = 1$ and $\text{cis } \frac{2\pi}{5}, \text{cis } (-\frac{2\pi}{5})$ and $\text{cis } \frac{4\pi}{5}, \text{cis } (-\frac{4\pi}{5})$ (arranged with conjugates paired). We may factor $x^5 - 1$ over the complex numbers;

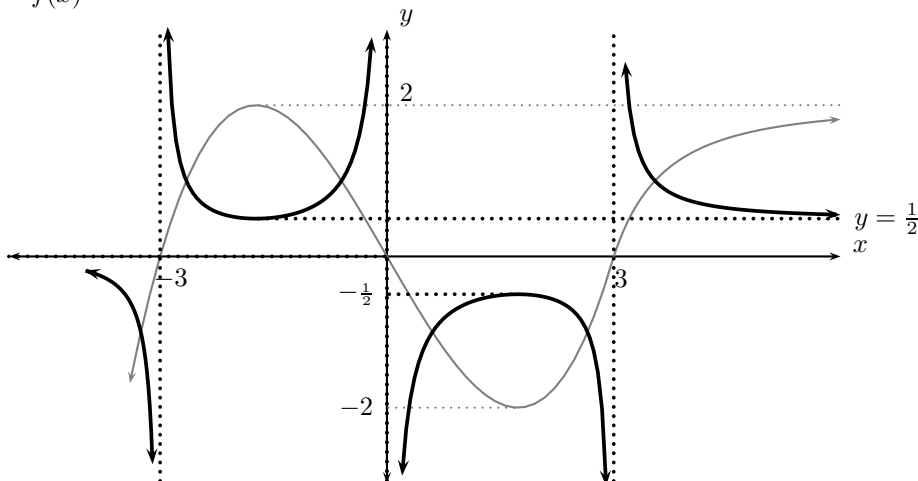
$$(x - 1)(x - \text{cis } \frac{2\pi}{5})(x - \text{cis } (-\frac{2\pi}{5}))(x - \text{cis } \frac{4\pi}{5})(x - \text{cis } (-\frac{4\pi}{5}))$$

multiplying out the conjugate pairs gives;

$$(x - 1)(x^2 - 2x \cos \frac{2\pi}{5} + 1)(x^2 - 2x \cos \frac{4\pi}{5} + 1)$$

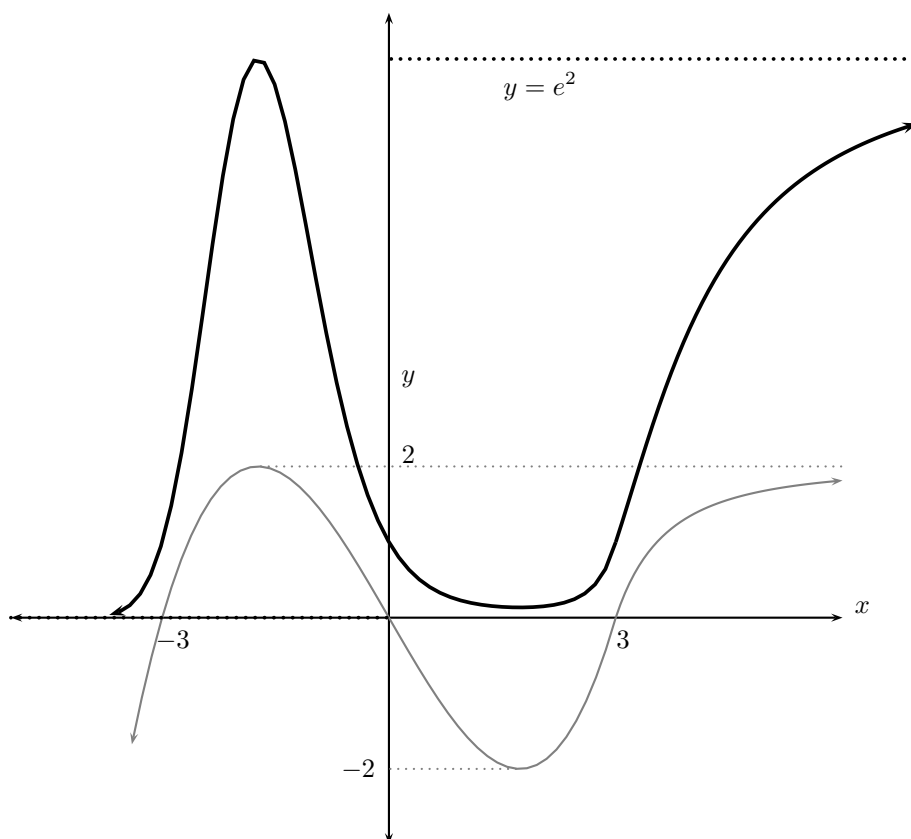
(b) (i) $y = \frac{1}{f(x)}$

2



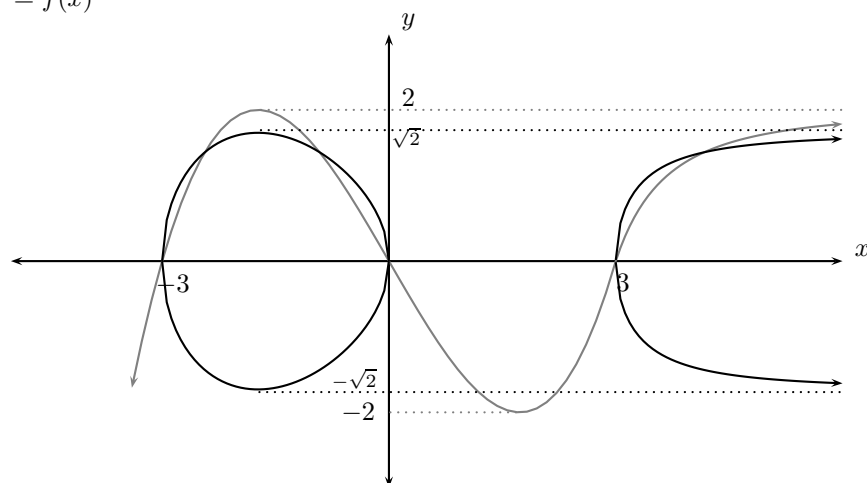
(ii) $y = e^{f(x)}$

3



(iii) $y^2 = f(x)$

3



(c)

(i) Solving these equations simultaneously to find the point of intersection gives;

$$9x^3 - 48x^2 + 55x - 12 = 3x + 4.$$

And rearranging gives the required equation;

$$P(x) = 9x^3 - 48x^2 + 52x - 16 = 0.$$

1

- (ii) At A the curves are tangential, which means that the equation will have a double root. This must be a single root of $P'(x) = 27x^2 - 96x + 52$. This has roots **3**

$$\begin{aligned} x &= \frac{96 \pm \sqrt{3600}}{54} \\ &= \frac{26}{9} \quad \text{or} \quad \frac{2}{3} \end{aligned}$$

Checking these roots in $P(x)$ we find only $x = \frac{2}{3}$ is a root of both the original polynomial *and* its derivative, so is the required x -coordinate of A . Since the roots of $P(x)$ multiply to $\frac{16}{9}$, the other root is

$$\frac{16}{9} \div \left(\frac{2}{3}\right)^2 = 4$$

QUESTION FIVE (15 marks)

Marks

- (a) The equation of this parabola is $y = -\frac{h}{a^2}(x^2 - a^2)$. **3**

$$\begin{aligned} \text{Area} &= \frac{2h}{a^2} \int_0^a (a^2 - x^2) dx \\ &= \frac{2h}{a^2} \left[a^2x - \frac{1}{3}x^3 \right]_0^a \\ &= \frac{2h}{a^2} \left(a^3 - \frac{1}{3}a^3 \right) \\ &= \frac{2h}{a^2} \times \frac{a^3}{3} \\ &= \frac{4}{3}ha \end{aligned}$$

- (b) (i) A slice taken at x has base running from $y = -\sqrt{1 - \frac{x^2}{4}}$ to $y = +\sqrt{1 - \frac{x^2}{4}}$. We can use part (a) **1**
with $a = \sqrt{1 - \frac{x^2}{4}}$ and $h = 2a$. Thus the area of a slice is

$$\begin{aligned} \frac{4}{3}ha &= \frac{4}{3} \times 2\sqrt{1 - \frac{x^2}{4}} \times \sqrt{1 - \frac{x^2}{4}} \\ &= \frac{8}{3} \left(1 - \frac{x^2}{4} \right) \\ &= \frac{8 - 2x^2}{3} \end{aligned}$$

- (ii) The volume required is **2**

$$\begin{aligned} V &= \lim_{\delta x \rightarrow 0} 2 \times \sum_{x=0}^2 \frac{8 - 2x^2}{3} \delta x \\ &= 2 \int_0^2 \frac{8 - 2x^2}{3} dx \\ &= \frac{2}{3} \left[8x - \frac{2}{3}x^3 \right]_0^2 \\ &= \frac{2}{3} \left(16 - \frac{16}{3} \right) \\ &= \frac{64}{9} \end{aligned}$$

- (c) (i) Using the binomial expansion on
- $(\cos \theta + i \sin \theta)^7$
- gives;

$$\begin{aligned} & \cos^7 \theta + 7 \cos^6 \theta i \sin \theta + 21 \cos^5 \theta i^2 \sin^2 \theta + 35 \cos^4 \theta i^3 \sin^3 \theta \\ & + 35 \cos^3 \theta i^4 \sin^4 \theta + 21 \cos^2 \theta i^5 \sin^5 \theta + 7 \cos^1 \theta i^6 \sin^6 \theta + i^7 \sin^7 \theta \end{aligned}$$

- (ii) Simplifying our result from part (i);

$$\begin{aligned} & \cos^7 \theta - 21 \cos^5 \theta \sin^2 \theta + 35 \cos^3 \theta i^4 \sin^4 \theta - 7 \cos^1 \theta \sin^6 \theta \\ & + i(7 \cos^6 \theta \sin \theta - 35 \cos^4 \theta \sin^3 \theta + 21 \cos^2 \theta \sin^5 \theta - \sin^7 \theta) \end{aligned}$$

By de Moivre's theorem this is $\cos 7\theta + i \sin 7\theta$, hence

$$\begin{aligned} \tan 7\theta &= \frac{\sin 7\theta}{\cos 7\theta} \\ &= \frac{7 \cos^6 \theta \sin \theta - 35 \cos^4 \theta \sin^3 \theta + 21 \cos^2 \theta \sin^5 \theta - \sin^7 \theta}{\cos^7 \theta - 21 \cos^5 \theta \sin^2 \theta + 35 \cos^3 \theta i^4 \sin^4 \theta - 7 \cos^1 \theta \sin^6 \theta} \end{aligned}$$

If we now divide top and bottom by $\cos^7 \theta$, we get the required expression;

$$\tan 7\theta = \frac{7 \tan \theta - 35 \tan^3 \theta + 21 \tan^5 \theta - \tan^7 \theta}{1 - 21 \tan^2 \theta + 35 \tan^4 \theta - 7 \tan^6 \theta}$$

- (iii) If we let
- $x = \tan \theta$
- then

$$\tan 7\theta = \frac{7x - 35x^3 + 21x^5 - x^7}{1 - 21x^2 + 35x^4 - 7x^6}$$

Thus the solutions of $\tan 7\theta = 0$ are the solutions of $7x - 35x^3 + 21x^5 - x^7 = 0$. This seventh degree polynomial factorises as $-x(x^6 - 21x^4 + 35x^2 - 7) = 0$. One solution is the trivial solution $x = 0$ and other other solutions will be the non-trivial solutions of $\tan 7\theta = 0$.

The solutions of $\tan 7\theta = 0$ are

$$7\theta = k\pi, \quad k \in \mathbf{Z}$$

seven distinct solutions are $\theta = 0, \frac{\pi}{7}, -\frac{\pi}{7}, \frac{2\pi}{7}, -\frac{2\pi}{7}, \frac{3\pi}{7}, -\frac{3\pi}{7}$. Thus the required roots of the degree six polynomials are

$$\pm \tan \frac{\pi}{7}, \quad \pm \tan \frac{2\pi}{7}, \quad \pm \tan \frac{3\pi}{7}$$

- (iv) The product of roots for the degree six polynomial in part (iii) is

$$-\tan^2 \frac{\pi}{7} \times -\tan^2 \frac{2\pi}{7} \times -\tan^2 \frac{3\pi}{7} = -\frac{7}{1}$$

Hence $\tan \frac{\pi}{7} \tan \frac{2\pi}{7} \tan \frac{3\pi}{7} = \sqrt{7}$ (we take the positive root since all the angles are first quadrant).

- (v) One way to tackle this problem is to notice that the
- cubic**
- in
- x^2

$$(x^2)^3 - 21(x^2)^2 + 35(x^2) - 7 = 0$$

has roots $\tan^2 \frac{\pi}{7}$, $\tan^2 \frac{2\pi}{7}$ and $\tan^2 \frac{3\pi}{7}$. If we label these roots α , β and γ then the required expression is

$$\begin{aligned} \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\ &= \frac{35}{+7} \\ &= 5 \end{aligned}$$

Alternative method: The polynomial with reciprocal roots is

$$\left(\frac{1}{x}\right)^6 - 21\left(\frac{1}{x}\right)^4 + 35\left(\frac{1}{x}\right)^2 - 7 = 0$$

i.e. $1 - 21x^2 + 35x^4 - 7x^6 = 0$. Regarding this as a cubic in x^2 , the required expression is the sum of the roots i.e. $-\frac{35}{-7} = 5$, as before.

QUESTION SIX (15 marks)**Marks**

- (a) (i) It is enough to show that the diagonals bisect, i.e that
- $PO = RO$
- and
- $QO = SO$
- .

2

$$\begin{aligned}
 RO^2 &= (-ct_1)^2 + \left(-\frac{c}{t_1}\right)^2 & QO^2 &= (-ct_2)^2 + \left(-\frac{c}{t_2}\right)^2 \\
 &= (ct_1)^2 + \left(\frac{c}{t_1}\right)^2 & &= (ct_2)^2 + \left(\frac{c}{t_2}\right)^2 \\
 &= PO^2 & &= SO^2
 \end{aligned}$$

Hence it is a parallelogram.

Alternative Method: Show that the midpoint of both PR and QS is O .**Alternative Method:** Consider gradients.

- (ii)

2

$$\begin{aligned}
 \text{The gradient at } P &= \frac{\dot{y}}{\dot{x}} \\
 &= \frac{-\frac{c}{t_1^2}}{c} \\
 &= -\frac{1}{t_1^2}
 \end{aligned}$$

$$\text{Hence } y - y_1 = m(x - x_1)$$

$$y - \frac{c}{t_1} = -\frac{1}{t_1^2}(x - ct_1)$$

$$t_1^2 y - ct_1 = -x + ct_1$$

$$x + t_1^2 y = 2ct_1$$

Hence we have the tangents:

$$\text{at } P: x + t_1^2 y = 2ct_1 \quad \dots\dots(1)$$

$$\text{at } Q: x + t_2^2 y = 2ct_2 \quad \dots\dots(2)$$

- (iii) We shall solve (1) and (2) simultaneously.

2

(1) - (2) gives

$$(t_1^2 - t_2^2)y = 2c(t_1 - t_2)$$

$$(t_1 + t_2)y = 2c$$

$$y = \frac{2c}{t_1 + t_2}$$

 $t_2^2(1) - t_1^2(2)$ gives

$$(t_2^2 - t_1^2)x = 2c(t_1 t_2^2 - t_1^2 t_2)$$

$$(t_2^2 - t_1^2)x = 2ct_1 t_2(t_2 - t_1)$$

$$(t_1 + t_2)x = 2ct_1 t_2$$

$$x = \frac{2ct_1 t_2}{t_1 + t_2}$$

- (iv) The coordinates of
- W
- ,
- X
- ,
- Y
- and
- Z
- are:

1

$$W\left(\frac{2ct_1 t_2}{t_1 + t_2}, \frac{2c}{t_1 + t_2}\right) \quad X\left(\frac{-2ct_1 t_2}{-t_1 + t_2}, \frac{2c}{-t_1 + t_2}\right)$$

$$Y\left(\frac{2ct_1 t_2}{-t_1 - t_2}, \frac{2c}{-t_1 - t_2}\right) \quad Z\left(\frac{-2ct_1 t_2}{t_1 - t_2}, \frac{2c}{t_1 - t_2}\right)$$

These expressions may be simplified to:

$$W\left(\frac{2ct_1 t_2}{t_1 + t_2}, \frac{2c}{t_1 + t_2}\right) \quad X\left(\frac{2ct_1 t_2}{t_1 - t_2}, \frac{-2c}{t_1 - t_2}\right)$$

$$Y\left(\frac{-2ct_1 t_2}{t_1 + t_2}, \frac{-2c}{t_1 + t_2}\right) \quad Z\left(\frac{-2ct_1 t_2}{t_1 - t_2}, \frac{2c}{t_1 - t_2}\right)$$

- (v) It is evident from the expressions above that the midpoint of both WY and XZ is the origin O .
Hence the diagonals bisect each other and the quadrilateral is a parallelogram.

1

(b) (i)

3

$$\begin{aligned}
 I_{-1} &= \int_1^2 \frac{(x-1)^{-\frac{1}{2}}}{x} dx \\
 &= \int_1^2 \frac{dx}{x\sqrt{x-1}} \\
 &= \int_0^1 \frac{2u du}{(u^2+1)u} \\
 &= \left[\tan^{-1} u \right]_0^1 \\
 &= \frac{\pi}{2}
 \end{aligned}
 \quad \begin{array}{l} \vdots \\ \text{Let } u^2 = x-1 \\ \text{Then } 2u du = dx \\ \text{and } x = u^2 + 1 \\ \text{and } \sqrt{x-1} = u \end{array}$$

(ii)

3

$$\begin{aligned}
 I_n &= \int_1^2 \frac{(x-1)(x-1)^{\frac{n}{2}-1}}{x} dx \\
 &= \int_1^2 \frac{x(x-1)^{\frac{n}{2}-1}}{x} dx - \int_1^2 \frac{(x-1)^{\frac{n}{2}-1}}{x} dx \\
 &= \left[\frac{2}{n} (x-1)^{\frac{n}{2}} \right]_1^2 - I_{n-2} \\
 &= \frac{2}{n} - I_{n-2}
 \end{aligned}$$

(iii)

1

$$\begin{aligned}
 I_5 &= \frac{2}{5} - I_3 \\
 &= \frac{2}{5} - \left(\frac{2}{3} - I_1 \right) \\
 &= \frac{2}{5} - \frac{2}{3} + I_1 \\
 &= \frac{2}{5} - \frac{2}{3} + \frac{2}{1} - I_{-1} \\
 &= \frac{26}{15} - \frac{\pi}{2}
 \end{aligned}$$

QUESTION SEVEN (15 marks)**Marks**

(a) (i)

1

$$\begin{aligned}
 m\ddot{x} &= -mg - \frac{mv^3}{100} \\
 v \frac{dv}{dx} &= -g - \frac{v^3}{100} \\
 v \frac{dv}{dx} &= \frac{-100g - v^3}{100} \\
 \frac{dv}{dx} &= \frac{100g + v^3}{-100v}
 \end{aligned}$$

(ii)

3

$$\begin{aligned}
 \frac{100v}{v^3 + 1000} &= \frac{A}{v + 10} + \frac{Bv + C}{v^2 - 10v + 100} \\
 100v &= A(v^2 - 10v + 100) + (Bv + C)(v + 10)
 \end{aligned}$$

Let $v = -10$: it follows that $-1000 = 300A + 0$ so $A = -\frac{10}{3}$.

Let $v = 0$: it follows that

$$\begin{aligned}
 0 &= 100A + 10C \\
 &= -\frac{1000}{3} + 10C
 \end{aligned}$$

$$\text{Hence } C = \frac{100}{3}.$$

Finally, equating coefficients of v^2 gives $0 = A + B$, hence $B = -A = \frac{10}{3}$.

(iii)

3

$$\begin{aligned}
 x &= \frac{10}{3} \int \frac{1}{v + 10} dv - \frac{10}{3} \int \frac{v + 10}{v^2 - 10v + 100} dv \\
 &= \frac{10}{3} \ln |v + 10| - \frac{10}{6} \int \frac{2v - 10}{v^2 - 10v + 100} dv - 50 \int \frac{dv}{v^2 - 10v + 100} \\
 &= \frac{10}{3} \ln |v + 10| - \frac{10}{6} \ln |v^2 - 10v + 100| - \frac{50}{5\sqrt{3}} \tan^{-1} \frac{v - 5}{5\sqrt{3}} + C
 \end{aligned}$$

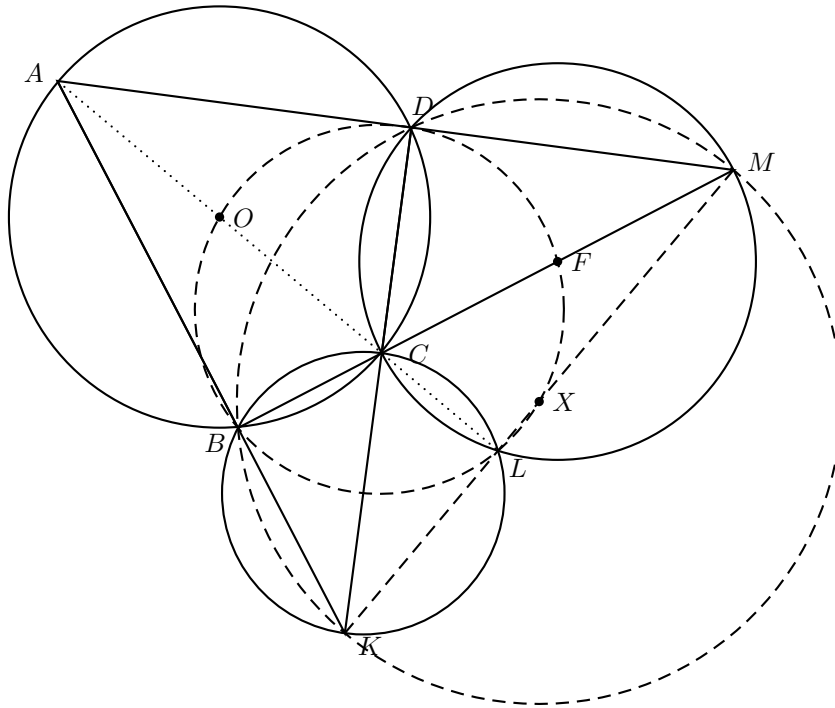
Using the initial condition $v = 5$ when $x = 0$,

$$0 = \frac{10}{3} \ln 15 - \frac{10}{6} \ln 75 - 0 + C$$

Hence $C = -\frac{5}{3} \ln \frac{15^2}{75} = -\frac{5}{3} \ln 3$. The maximum height is obtained when $v = 0$, hence

$$\begin{aligned}
 \text{Maximum height} &= \frac{10}{3} \ln 10 - \frac{5}{3} \ln 100 - \frac{10}{\sqrt{3}} \tan^{-1} \left(-\frac{1}{\sqrt{3}} \right) - \frac{5}{3} \ln 3 \\
 &= -\frac{5}{3} \ln 3 + \frac{10}{6\sqrt{3}} \pi \\
 &\doteq 1.19 \text{ metres}
 \end{aligned}$$

(b)



- (i) $\angle CDA = 90^\circ$ (angle in a semicircle) and so $\angle CDM = 90^\circ$ (straight angle). Thus CM is the diameter of circle CDM and the centre F is the midpoint of CM . 1

(ii)

$$\angle KLC = \angle CBA \quad (\text{exterior opposite angle of cyclic quadrilateral } KLCB)$$

$$= 90^\circ \quad (\text{angle in semicircle})$$

$$\angle MLC = \angle CDA \quad (\text{exterior opposite angle of cyclic quadrilateral } MLCD)$$

$$= 90^\circ \quad (\text{angle in semicircle})$$

Hence $\angle KLC + \angle MLC = 180^\circ$ and so KLM is straight.

- (iii) We have already shown that $\angle CBA = 90^\circ$. Hence $\angle KBM = 90^\circ$ (straight angle). Thus KM is the diameter of a circle through KMB (converse to angle in a semicircle theorem). 2

Similarly, $\angle KDM = 90^\circ$ (straight angle) and KM is the diameter of a circle through KMD . This must be the same circle, since they have a common diameter.

The centre X of this circle is the midpoint of KM .

(iv)

$$\angle DFC = 2\alpha \quad (\text{angle at centre and circumference on arc } DC)$$

$$\angle DXB = 2\alpha \quad (\text{angle at centre and circumference on arc } DB \text{ of circle } DBKM)$$

Since DB subtends equal angles at F and X there must be a circle through $DBXF$. 3

QUESTION EIGHT (15 marks)**Marks**

(a) (i) Expanding,

1

$$\begin{aligned} 8(z-2)^3 - (z-i)^3 &= 8(z^3 - 6z^2 + 12z - 8) - (z^3 - 3iz^2 + 3i^2z - i^3) \\ &= (8-1)z^3 + z^2(-48+3i) + z(96+3) - 64 - i \end{aligned}$$

(ii)

4

$$8(z-2)^3 - (z-i)^3 = 0$$

$$8(z-2)^3 = (z-i)^3$$

$$\left(\frac{2z-4}{z-i}\right)^3 = 1$$

Hence $\frac{2z-4}{z-i} = 1, \omega$ or ω^2 .The first solution: $\frac{2z-4}{z-i} = 1$

$$2z-4 = z-i$$

$$z = 4-i$$

The second solution: $\frac{2z-4}{z-i} = \omega$

$$2z-4 = \omega z - \omega i$$

$$z = \frac{4-i\omega}{2-\omega}$$

$$z = \frac{(4-i\omega)(2-\omega^2)}{(2-\omega)(2-\omega^2)}$$

$$z = \frac{(8+i-2i\omega-4\omega^2)}{4+1+2}$$

where we have used the identities $\omega + \omega^2 = -1$ and $\omega \times \omega^2 = \omega^3 = 1$. Since $\omega = \frac{1}{2} + i\frac{\sqrt{3}}{2}$ and $\omega^2 = \frac{1}{2} - i\frac{\sqrt{3}}{2}$ the second solution simplifies to

$$\frac{8+2+2i\sqrt{3}+i+\sqrt{3}+i}{7} = \frac{10+\sqrt{3}+2i(\sqrt{3}+1)}{7}$$

The calculation for the third solution is identical except ω and ω^2 are swapped, leading to

$$\begin{aligned} z &= \frac{(8+i-2i\omega^2-4\omega)}{7} \\ &= \frac{8+2-2i\sqrt{3}+i-\sqrt{3}+i}{7} \\ &= \frac{10-\sqrt{3}+2i(1-\sqrt{3})}{7} \end{aligned}$$

(The second and third solutions are like conjugates in the sense of swapping $\sqrt{3}$ and $-\sqrt{3}$.)

(b) (i)

4When $n = 0$,

$$\frac{a_n}{b_n} = \frac{1}{\sqrt{1+\tan^2\theta}}$$

and

$$\frac{\sin\theta}{b_n} = \frac{\sin\theta}{1} \quad (\text{as required})$$

$$= \frac{1}{\sqrt{\sec^2\theta}}$$

$$= \cos\theta \quad (\text{as required})$$

Assume the result holds for $n = k$. That is, assume that

$$\frac{a_k}{b_k} = \cos\frac{\theta}{2^k} \quad \text{and} \quad \frac{\sin\theta}{b_k} = 2^k \sin\frac{\theta}{2^k}.$$

We need to show that the result holds for $n = k + 1$, that is to prove that

$$\frac{a_{k+1}}{b_{k+1}} = \cos \frac{\theta}{2^{k+1}} \quad \text{and} \quad \frac{\sin \theta}{b_{k+1}} = 2^{k+1} \sin \frac{\theta}{2^{k+1}}.$$

Consider the LHS of the first expression:

$$\begin{aligned} \frac{a_{k+1}}{b_{k+1}} &= \frac{a_{k+1}}{\sqrt{a_{k+1}b_k}} \\ &= \sqrt{\frac{a_{k+1}}{b_k}} \\ &= \sqrt{\frac{\frac{1}{2}(a_k + b_k)}{b_k}} \\ &= \sqrt{\frac{1}{2}\left(\frac{a_k}{b_k} + 1\right)} \\ &= \sqrt{\frac{1}{2}\left(1 + \cos \frac{\theta}{2^k}\right)} \quad (\text{By the inductive assumption}) \\ &= \sqrt{\cos^2 \frac{1}{2} \frac{\theta}{2^k}} \quad (\text{By a double angle formula for } \cos) \\ &= \cos \frac{\theta}{2^{k+1}} \quad (\star\star) \end{aligned}$$

as required. For the second result, notice the recurrence relation $b_{k+1} = \sqrt{a_{k+1}b_k}$ may be rewritten $\frac{b_{k+1}}{b_k} = \frac{a_{k+1}}{b_{k+1}}$. Now consider the LHS of the second expression:

$$\begin{aligned} \frac{\sin \theta}{b_{k+1}} &= \frac{\sin \theta}{b_k} \times \frac{b_k}{b_{k+1}} \\ &= 2^k \sin \frac{\theta}{2^k} \times \frac{b_k}{b_{k+1}} \quad (\text{By the inductive assumption}) \\ &= 2^k \sin \frac{\theta}{2^k} \times \frac{b_{k+1}}{a_{k+1}} \\ &= 2^k \sin \frac{\theta}{2^k} \times \sec \frac{\theta}{2^{k+1}} \quad (\text{By } \star\star) \\ &= 2^k \times 2 \sin \left(\frac{1}{2} \frac{\theta}{2^k}\right) \cos \left(\frac{1}{2} \frac{\theta}{2^k}\right) \times \sec \frac{\theta}{2^{k+1}} \quad (\text{by the double angle formula for } \sin) \\ &= 2^{k+1} \sin \frac{\theta}{2^{k+1}} \end{aligned}$$

as required.

Hence the two results hold for all integers $n \geq 0$ by mathematical induction.

(ii)

$$\begin{aligned} \frac{\sin \theta}{b_n} &= 2^n \sin \frac{\theta}{2^n} \\ &= \theta \times \frac{\sin \frac{\theta}{2^n}}{\frac{\theta}{2^n}} \\ &\rightarrow \theta \times 1 \quad (\text{as } \frac{\theta}{2^n} \rightarrow 0, \text{ i.e. as } n \rightarrow \infty) \end{aligned}$$

$$\text{Hence } \lim_{n \rightarrow \infty} \frac{x}{b_n \sqrt{1+x^2}} = \tan^{-1} x, \text{ since } \frac{x}{\sqrt{1+x^2}} = \sin \theta.$$

(iii)

n	0	1	2	3
a_n	0.7071	0.8536	0.8887	0.8974
b_n	1	0.9239	0.9061	0.9018

2

1

(iv) Hence $\frac{\pi}{4} \doteq \frac{1}{b_3\sqrt{2}} \doteq 0.7851$. (Compare this with $\frac{\pi}{4} \doteq 0.785398$ using the calculators value of π .)

1

(v) Notice again that

2

$$\frac{b_{n+1}}{b_n} = \frac{a_{n+1}}{b_{n+1}} = \cos \frac{\theta}{2^{n+1}}$$

$$\begin{aligned} \text{Hence } 0 < 1 - \frac{b_{n+1}}{b_n} \\ &= 1 - \cos \frac{\theta}{2^{n+1}} \\ &= 2 \sin^2 \frac{\theta}{2^{n+2}} \end{aligned}$$

$$\text{Hence we require } 2 \sin^2 \frac{\theta}{2^{n+2}} < 10^{-10}$$

$$\sin \frac{\theta}{2^{n+2}} < \frac{10^{-5}}{\sqrt{2}}$$

$$\frac{\theta}{2^{n+2}} < \sin^{-1} \frac{10^{-5}}{\sqrt{2}}$$

$$2^{n+2} > \frac{\pi}{4 \sin^{-1} \frac{10^{-5}}{\sqrt{2}}}$$

$$n > \log_2 \frac{\pi}{4 \sin^{-1} \frac{10^{-5}}{\sqrt{2}}} - 2$$

$$n > 14.8 \quad (\text{approx})$$

Thus 15 iterations will give the desired accuracy.

BDD